

Report on Crime Data

Dataset: crime_data_bangladesh.csv, sourced from the Kaggle. The dataset contains crime and recovery case statistics across different regions of Bangladesh, categorized by year and area. It includes various crime types (e.g., violent, property crimes) and recovery cases (e.g., arms, explosives, narcotics) to analyze trends and regional patterns.

Data Cleaning and Preparation:

- **Removing N/A values:** No empty values observed in any column.
- **Removing Duplicates:** No duplicate values observed.
- **Standardizing Names of Column & Values:** Column names and Area names are standardized.
- **New Column Added:**
 - Total Cases: Calculates total cases per row.
 - Violent_Crimes: Total cases of violent cases, such as- murder, women/children repression.
 - Property_crimes: Total cases of property cases, such as- dacoity, robbery.
 - Sum_recovery_cases: Total cases of recovery cases.
 - violent_crime_ex_women_oppression: Total cases of violent crimes except Women/Children repression for calculation purpose.
- **Aggregated Population Data:** Population data¹ hardcoded to the Measure- *Adjusted Crime Rate*, which depicts crimes per 1000 people per year.

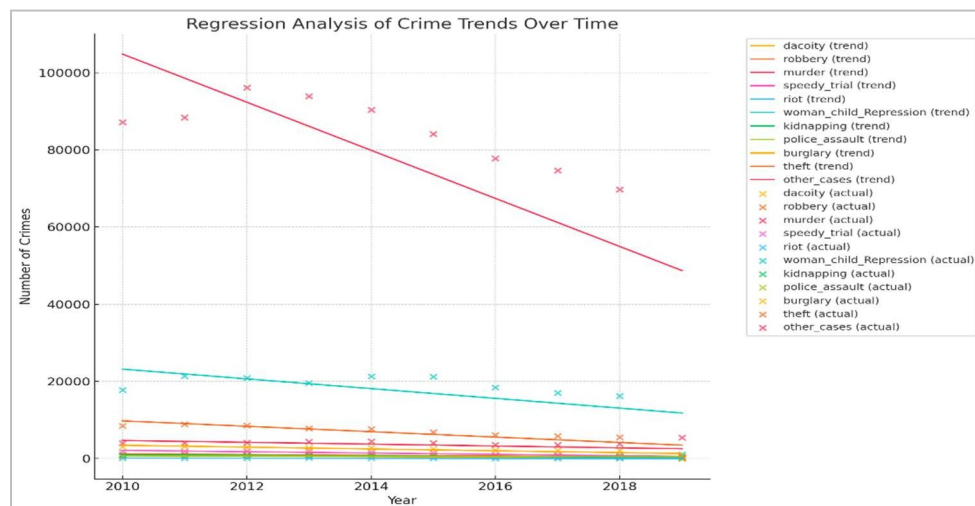
Basic Statistical Analysis:

	mean	median	std
dacoity	36.38776	17	66.82193
robbery	63.05442	30	103.45
murder	274.2925	165	434.8545
speedy_trial	104.2857	52	175.0705
riot	6.190476	1	13.56634
woman_child_Repression	1311.925	745	1934.303
kidnapping	51.86395	21	85.7861
police_assault	45.73469	21	65.55363
burglary	183.7823	79	306.9374
theft	509.4558	262	854.4761

¹ <https://www.macrotrends.net/global-metrics/countries/bgd/bangladesh/population>

- **High and Low Crime Types:**
 - The area with the **highest crime rate** is **Dhaka Division** with a total of 241,247 crimes.
 - The area with the **lowest crime rate** is **Rangpur Metropolitan** with a total of 199 crimes.
- **Regional Disparities:**
 - **Dhaka Division** has the highest total crimes, likely due to its large population, urban density, and higher activity levels.
 - **Rangpur Metropolitan** has the lowest total crimes, indicating potentially better crime control, lower population density, or less reported crime.

Trend Analysis using Regression:

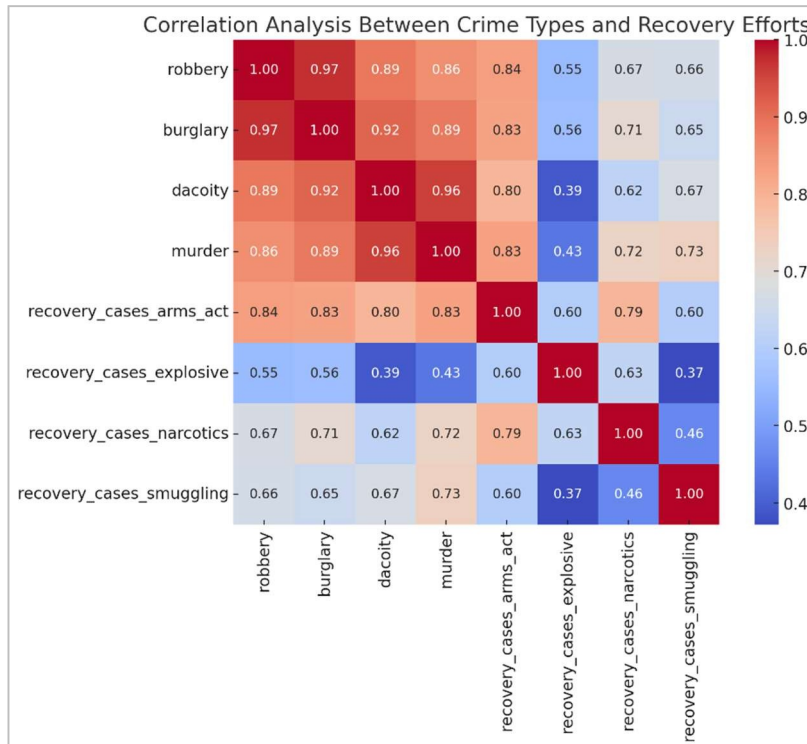


Observations:

- **General Decline:** All crime types show decreasing trends, highlighting overall improvements in crime management or underreporting over time.
- **Focus Areas:** Despite decreases, crimes like theft and crimes against women/children may still require focus due to their historically high prevalence.

Correlation Analysis:

Relationships between types of crimes (e.g., robbery and burglary) or between crimes and recovery efforts.



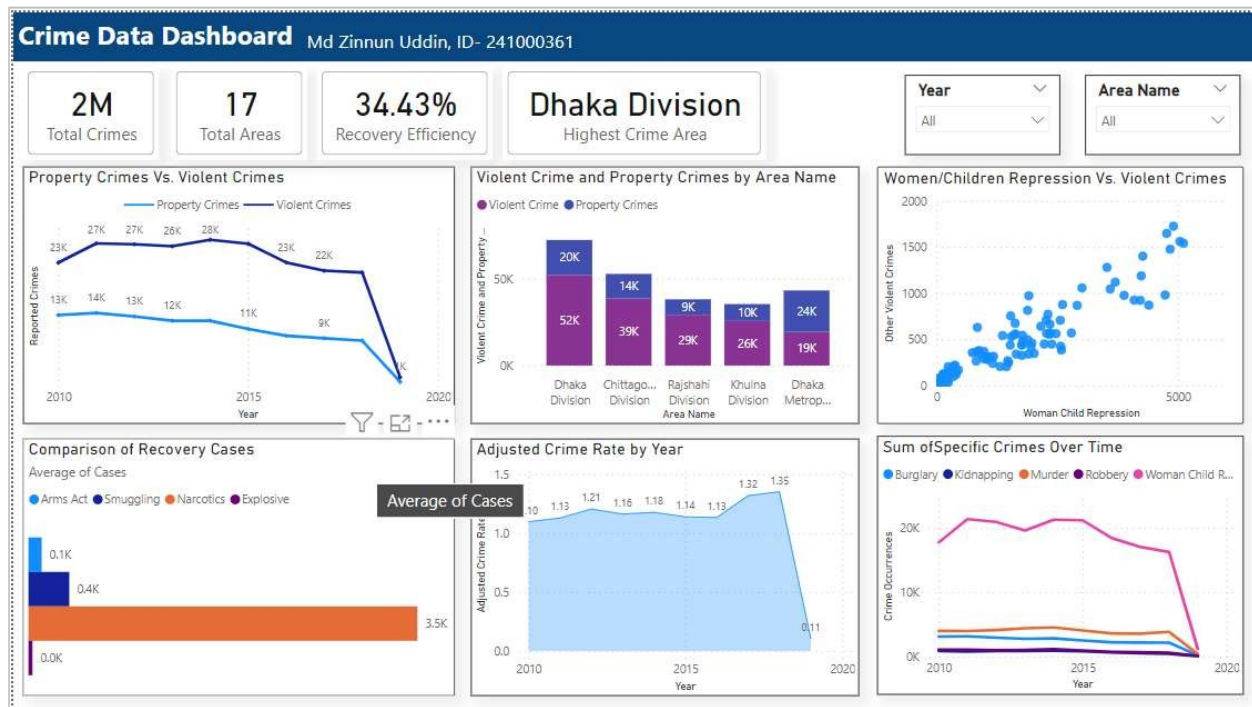
Key Observations:

- **Property Crimes (Robbery, Burglary, Dacoity)** show strong internal correlations and linkages with arms-related recoveries, highlighting a focus area for law enforcement.
- **Violent Crimes (Murder)** have significant relationships with narcotics and smuggling recoveries, suggesting a linkup between these activities.
- Recovery efforts for **arms, explosives, and narcotics** are interconnected, possibly reflecting shared enforcement strategies.

Purpose and Utilization of Dashboard:

The dashboard provides a comprehensive overview of crime and recovery trends across regions and years, enabling stakeholders to analyze patterns and assess law enforcement efficiency. It highlights key metrics like total crimes, recovery efficiency, and adjusted crime rates while visualizing property and violent crimes, recovery case comparisons, and correlations between crimes. Interactive filters for year and area allow granular analysis, helping policymakers and law enforcement focus resources on high-crime areas and improve recovery strategies.

Dashboard



Key Insights of Visualizations:

1. Cards (Metrics and Summary):

- Total Crimes (2M):** Displays the total number of crimes recorded in the dataset across all years and regions.
- Total Areas (17):** Shows the total number of distinct regions or areas where crimes were reported.
- Recovery Efficiency (34.43%):** Represents the percentage of recovery cases (e.g., arms, narcotics) relative to the total crimes.
- Highest Crime Area (Dhaka Division):** Highlights the region with the highest number of reported crimes, which is Dhaka Division.

2. Property Crimes vs. Violent Crimes Over Time:

- A line chart comparing trends in property crimes and violent crimes from 2010 to 2020.
- Observation:**
 - Property crimes remain consistently higher than violent crimes, though both decrease significantly after 2015.
 - A sharp decline in both crime categories is observed near 2020.

3. Violent Crimes and Property Crimes by Area Name:

- **Observation:**
 - Dhaka Division reports the highest number of both violent (52K) and property crimes (20K), followed by Chittagong Division.
 - Smaller regions like Khulna Division have comparatively lower crime numbers.

4. Women/Children Repression vs. Violent Crimes (Scatter Plot):

- **Observation:**
 - There is a strong positive correlation, indicating areas with higher women/children repression cases also report higher violent crimes.
 - Points clustered in the higher range suggest hotspot areas for both crime types.

5. Comparison of Recovery Cases:

- **Observation:**
 - Narcotics cases dominate with the highest average recovery (3.5K), while explosives and arms-related recoveries are significantly lower.

6. Adjusted Crime Rate by Year:

- An area chart showing the crime rate normalized by population from 2010 to 2020.
- **Observation:**
 - The adjusted crime rate remains steady around 1.1–1.3 crimes per 1,000 people until 2020, where it sharply drops.

7. Sum of Specific Crimes Over Time:

- A line chart visualizing trends in specific crime types (e.g., burglary, kidnapping, murder, robbery, and crimes against women/children) from 2010 to 2020.
- **Observation:**
 - Crimes against women/children show the highest numbers but decrease significantly after 2015.
 - Other crime types like burglary and robbery have relatively stable or declining trends.

Actionable Recommendations:

1. **Target High-Crime Areas:** Increase police presence, surveillance, and community engagement in Dhaka Division and other hotspots.
2. **Address Crimes Against Women/Children:** Deploy specialized task forces, raise awareness, and fast-track legal processes.
3. **Enhance Recovery Efforts:** Focus on arms, narcotics, and smuggling recoveries through better intelligence and interagency cooperation.
4. **Leverage Technology:** Use predictive analytics, GIS mapping, and real-time dashboards for proactive crime management.
5. **Strengthen Public Awareness:** Educate communities on crime prevention, address socio-economic issues, and engage youth in positive activities.
6. **Adjust Resource Allocation:** Focus on areas with high adjusted crime rates and conduct annual audits to refine strategies.
7. **Improve Reporting Mechanisms:** Simplify and anonymize reporting processes, especially for sensitive crimes.

These actions can help law enforcement effectively reduce crime rates and improve recovery efficiency.

Summary of Findings and Value of Analysis

The analysis highlights key crime trends, with Dhaka Division identified as the highest-crime area, dominated by property and violent crimes. Crimes against women and children show strong correlations with other violent crimes, requiring targeted interventions. While overall crime rates and recovery efficiency indicate progress, certain categories like narcotics-related recoveries remain a priority.

The value of this analysis lies in its actionable insights, enabling law enforcement to allocate resources effectively, focus on high-crime regions, and enhance recovery efforts. By leveraging trends, correlations, and adjusted crime rates, policymakers can design data-driven strategies to improve public safety and reduce crime sustainably.